Math 550, S20
Exam 1

Name: Answer

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	21	
2	21	
3	21	
4	21	
5	20	
Total	104	

Good Luck !

Problem 1. (21 points) Given two vectors $\mathbf{a} = \sqrt{3}\,\mathbf{i} + 3\,\mathbf{j} + 2\sqrt{6}\,\mathbf{k}$ and $\mathbf{b} = (-\sqrt{3},\ 0,\ \sqrt{6})$.
(a) Find $\|\mathbf{a}\|$, $\|\mathbf{b}\|$, and the unit vector $\mathbf{u}$ which is in the same direction as $\mathbf{a}$.

$$\|a\| = \sqrt{(\sqrt{3})^2 + 3^2 + (2\sqrt{6})^2} = \sqrt{3+9+24} = \sqrt{36} = 6$$

$$\|b\| = \sqrt{(-\sqrt{3})^2 + 0^2 + (\sqrt{6})^2} = \sqrt{3+0+6} = \sqrt{9} = 3$$

$$u = \frac{a}{\|a\|} = \frac{1}{6}(\sqrt{3}, 3, 2\sqrt{6}) = \left(\frac{\sqrt{3}}{6}, \frac{1}{2}, \frac{\sqrt{6}}{3}\right)$$

(c) Find the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

$$a \cdot b = (\sqrt{3}, 3, 2\sqrt{6}) \cdot (-\sqrt{3}, 0, \sqrt{6})$$

$$= (\sqrt{3})(-\sqrt{3}) + 3 \times 0 + 2\sqrt{6} \times \sqrt{6} = -3 + 0 + 12 = 9$$

$$\Rightarrow \cos\theta = \frac{a \cdot b}{\|a\|\,\|b\|} = \frac{9}{6 \cdot 3} = \frac{9}{18} = \frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ or } 60^\circ$$

(d) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, $\mathrm{proj}_{\mathbf{a}}\mathbf{b}$.

$$\mathrm{Proj}_a b = \left(\frac{a \cdot b}{\|a\|^2}\right) a$$

$$= \frac{9}{6^2}(\sqrt{3}, 3, 2\sqrt{6})$$

$$= \left(\frac{\sqrt{3}}{4}, \frac{3}{4}, \frac{\sqrt{6}}{2}\right)$$

Problem 2. (21 points) Given three points $P(-1,3,1)$, $Q(0,5,-2)$, and $R(4,3,-1)$ in $\mathbb{R}^3$.
(a) Find parametric equations for the line through the two points P and Q.

$$\overrightarrow{PQ} = (0,5,-2)-(-1,3,1) = (1,2,-3)$$

$$\Rightarrow \quad r(t) = (-1,3,1)+t(1,2,-3)$$

i.e., $\begin{cases} x(t) = -1+t, \\ y(t) = 3+2t, \\ z(t) = 1-3t, \end{cases} \quad -\infty < t < +\infty$

(b) Find an equation for the plane through the point P normal to the vector $2\,\mathbf{i}-\mathbf{j}+3\,\mathbf{k}$.

$$\vec{n} = 2i-j+3k = (2,-1,3), \quad P=(-1,3,1)$$

$$2(x+1)-(y-3)+3(z-1)=0$$

$$2x+2-y+3+3z-3=0$$

$$2x-y+3z=-2$$

(c) Find an equation for the plane that contains the triangle $\triangle PQR$.

$$\overrightarrow{PQ}=(1,2,-3), \quad \overrightarrow{PR}=(5,0,-2)$$

$$\vec{n} = \overrightarrow{PQ}\times\overrightarrow{PR} = \begin{vmatrix} i & j & k \\ 1 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = i\begin{vmatrix} 2 & -3 \\ 0 & -2 \end{vmatrix} - j\begin{vmatrix} 1 & -3 \\ 5 & -2 \end{vmatrix} + k\begin{vmatrix} 1 & 2 \\ 5 & 0 \end{vmatrix}$$

$$= -4i-13j-10k$$

Thus we have the equation.

$$-4(x+1)-13(y-3)-10(z-1)=0$$

$$4x+13y+10z=45$$

Problem 3. (21 points)

(a) Compute $\lim\limits_{(x,y)\to(1,2)} \dfrac{4x^2-y^2}{2x-y}$

$$= \lim_{(x,y)\to(1,2)} \frac{(2x-y)(2x+y)}{2x-y} = \lim_{(x,y)\to(1,2)} 2x+y$$

$$= 2\cdot 1+2 = 4$$

(b) Use the **Chain Rule** to find $D(f\circ \mathbf{g})$ when $f(x,y) = ye^{x^2}$ and $\mathbf{g}(s,t) = (st, s+t)$.

$$Df = [2xye^{x^2} \quad e^{x^2}], \quad Dg = \begin{bmatrix} t & s \\ 1 & 1 \end{bmatrix}, \quad (x = st,\ y = s+t)$$

$$\Rightarrow D(f\circ g) = Df\,Dg = [2xye^{x^2} \quad e^{x^2}]\begin{bmatrix} t & s \\ 1 & 1 \end{bmatrix}$$

$$= [2xye^{x^2}t + e^{x^2} \quad 2xye^{x^2}s + e^{x^2}]$$

$$= e^{x^2}[2xyt+1 \quad 2xys+1]$$

$$= e^{s^2t^2}[2st^2(s+t)+1 \quad 2s^2t(s+t)+1]$$

(c) Find an equation for the tangent plane to the paraboloid $x^2 + \dfrac{y^2}{4} + \dfrac{z^2}{9} = 3$ at the point $(1, 2, 3)$.

$$f(x,y,z) = x^2 + \frac{y^2}{4} + \frac{z^2}{9} - 3$$

$$\Rightarrow \nabla f(x,y,z) = (2x, \tfrac{y}{2}, \tfrac{2}{9}z)$$

and $\nabla f(1,2,3) = (2, 1, \tfrac{2}{3}) \to$ the normal of the plane

The tangent plane is then given by

$$2(x-1) + (y-2) + \tfrac{2}{3}(z-3) = 0$$

$$2x + y + \tfrac{2}{3}z = 6$$

Problem 4. (21 points) Let $\mathbf{x}(t) = (e^t \cos t,\ e^t \sin t,\ 1)$ that defines a path in $\mathbb{R}^3$.
(a) Find the unit tangent vector $\mathbf{T}$ of the path.

$$x'(t) = (e^t\cos t - e^t\sin t,\ e^t\sin t + e^t\cos t,\ 0)$$
$$= (e^t(\cos t - \sin t),\ e^t(\sin t + \cos t),\ 0)$$
$$\|x'(t)\| = \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 + 0^2} \qquad \cos^2 t + \sin^2 t = 1$$
$$= \sqrt{e^{2t}(1 - 2\cos t\sin t) + e^{2t}(1 + 2\cos t\sin t)}$$
$$= \sqrt{2e^{2t}} = \sqrt{2}e^t$$
$$\Rightarrow T(t) = \frac{x'(t)}{\|x'(t)\|} = \frac{1}{\sqrt{2}e^t}(e^t(\cos t - \sin t),\ e^t(\sin t + \cos t),\ 0)$$
$$= \frac{1}{\sqrt{2}}(\cos t - \sin t,\ \sin t + \cos t,\ 0)$$

(b) Find the length of the arc defined by the path with $0 \le t \le 1$.

$$L = \int_0^1 \|x'(t)\|\,dt = \int_0^1 \sqrt{2}e^t\,dt$$
$$= \sqrt{2}e^t\Big|_0^1 = \sqrt{2}(e-1)$$

(c) Find the curvature κ and the principal unit normal vector $\mathbf{N}$ of the path.

$$T'(t) = \frac{1}{\sqrt{2}}(-\sin t - \cos t,\ \cos t - \sin t,\ 0)$$
$$\|T'(t)\| = \frac{1}{\sqrt{2}}\sqrt{(-\sin t - \cos t)^2 + (\cos t - \sin t)^2}$$
$$= \frac{1}{\sqrt{2}}\sqrt{(1 + 2\sin t\cos t) + (1 - 2\cos t\sin t)}$$
$$= \frac{1}{\sqrt{2}} \times \sqrt{2} = 1$$
$$\Rightarrow \kappa(t) = \frac{\|T'(t)\|}{\|x'(t)\|} = \frac{1}{\sqrt{2}e^t} = \frac{\sqrt{2}}{2}e^{-t}$$
$$N(t) = \frac{T'(t)}{\|T'(t)\|} = \frac{1}{\sqrt{2}}(-\sin t - \cos t,\ \cos t - \sin t,\ 0)$$

Problem 5. (20 points)

(a) Compute the divergence and the curl of the vector field $\mathbf{F} = y^2 z\,\mathbf{i} + e^{2xyz}\,\mathbf{j} + x^2 y\,\mathbf{k}$.

$$\nabla \cdot F = \frac{\partial}{\partial x}(y^2 z) + \frac{\partial}{\partial y}(e^{2xyz}) + \frac{\partial}{\partial z}(x^2 y) = 2xze^{2xyz}$$

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 z & e^{2xyz} & x^2 y \end{vmatrix} = i\begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{2xyz} & x^2 y \end{vmatrix} - j\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ y^2 z & x^2 y \end{vmatrix} + k\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ y^2 z & e^{2xyz} \end{vmatrix}$$

$$= i(x^2 - 2xye^{2xyz}) - j(2xy - y^2) + k(2yze^{2xyz} - 2yz)$$

$$= x(x - 2ye^{2xy})i + y(y - 2x)j + 2yz(e^{2xy} - 1)k$$

(b) Find the second order Taylor polynomial $P_2(x, y, z)$ for the function $f(x, y, z) = \dfrac{1}{x^2 + y^2 + z^2 + 1}$ at the point $\mathbf{a} = (0, 0, 0)$.

Let $r = x^2 + y^2 + z^2 + 1$, then

$$Df = \left[\frac{-2x}{r^2} \quad \frac{-2y}{r^2} \quad \frac{-2z}{r^2} \right] \Rightarrow Df(0,0,0) = [0 \quad 0 \quad 0]$$

$$Hf = \begin{bmatrix} \frac{-2}{r^2} + \frac{8x^2}{r^3} & \frac{8xy}{r^3} & \frac{8xz}{r^3} \\ \frac{8yx}{r^3} & \frac{-2}{r^2} + \frac{8y^2}{r^3} & \frac{8yz}{r^3} \\ \frac{8zx}{r^3} & \frac{8zy}{r^3} & \frac{-2}{r^2} + \frac{8z^2}{r^3} \end{bmatrix} \Rightarrow Hf(0,0,0) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

Thus

$$P_2(x,y,z) = f(a) + Df(a)(x-a) + \tfrac{1}{2}(x-a)^T Hf(a)(x-a)$$

$$= 1 + [0 \quad 0 \quad 0]\begin{bmatrix} x \\ y \\ z \end{bmatrix} + \tfrac{1}{2}[x \quad y \quad z]\begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= 1 - x^2 - y^2 - z^2$$